

Education

- 2022–2025 **Doctoral Training, Geometry and Dynamics of Distributed Cortical Sensory Representations**, INCC, Université Paris Cité, Paris
 - Cognitive Neuroscience, specialization in Computational and Theoretical Neuroscience
- 2022–2025 **Visiting Researcher**, Nuffield Department of Clinical Neurosciences, University of Oxford, Oxford
 - Research project: Cortical gradients of functional integration
- 2019–2021 **Predoctoral Researcher**, Center for Neural Science, NYU, New York City
 - Theoretical Neuroscience and Dynamical Systems
- 2018–2019 **Master of Science**, CogMaster, École normale supérieure, Paris
 - Cognitive Neuroscience
- 2010–2014 **Bachelor & Master in Engineering**, Telecom Paris, Paris, GPA – 3.84
 - Artificial Intelligence & Entrepreneurship
- Spring 2013 **Undergraduate**, Columbia University, New York City
 - Theoretical Neuroscience
- 2007–2010 **Undergraduate**, Lycée Charlemagne, Paris
 - Mathematics, Physics, Computer Science (MP*)

Academic Experiences

- 2026–Present **Postdoctoral Researcher**, Université de Montréal, Montréal, Geometry-Constrained Artificial Neural Networks for Sustainable AI and Personalized Brain Modeling
 - Investigating how biologically inspired geometric constraints on artificial neural networks can improve learning stability, computational efficiency, and energy efficiency.
 - Developing neural network architectures embedded on cortical-like surfaces, with predominantly local connectivity complemented by a small number of strategic long-range connections.
 - Applying these models to patient-specific brain simulations to study seizure propagation and support personalized intervention strategies in epilepsy.
 - Modeling tools: PyTorch, geometric deep learning, brain network modeling
 - Supervised by: Guillaume Dumas, Alexander Weil
- 2025 **Postdoctoral Researcher**, Freie Universität Berlin, Berlin, Emergence of Macro and Meso-scale Cortical Architecture in Embedded ANNs
 - Preliminary investigation of how large-scale cortical organization can emerge spontaneously within artificial neural networks embedded on cortical geometry and bound by biologically inspired connectivity rules.
 - Combination of reward-based learning and Hebbian plasticity to study the joint shaping of functional specialization.
 - Modeling tools: PyTorch, custom geometric deep learning framework.
 - Supervised by: Adrien Doerig

- 2022–2025 **PhD Candidate**, *Université Paris Cité*, Paris, Macro-scale Gradients of Cortical Functional Organization
- Investigated the competition for cortical territory across sensory systems, revealing the inhibitory role of the Default Mode Network in shaping functional specialization.
 - Developed a novel framework - Spatial Component Decomposition - to dissect the spatial influences shaping cortical maps using sparse dictionary learning on geodesic manifolds.
 - Integrated five large-scale fMRI datasets spanning three modalities and over 400 participants to ensure cross-modal generalizability.
 - Modeling tools: FreeSurfer, Nilearn, homemade machine learning package.
 - Supervised by: Daniel Margulies, Sean Froudish-Walsh
 - Klatzmann, U. et al. Spatial layout of visual specialization is shaped by competing default mode and sensory networks, *preprint* 10.1101/2025.09.08.674858 (2025)
- 2019–2021 **Predoctoral Researcher**, *NYU*, New York, Modeling Global Neuronal Workspace within Large-Scale Networks
- Implemented the Global Neuronal Workspace architecture proposed by Stanislas Dehaene and colleagues within the large-scale framework developed by Xiao-Jing Wang and colleagues, based on the macaque cortex architecture (Kennedy and colleagues). Explored the impact of AMPA, NMDA and GABA gating variable ratios on local and global dynamics.
 - Modeling tools: Brian and PyDSTool
 - Supervised by: Xiao-Jing Wang & Sean Froudish-Walsh
 - Collaborators: Stanislas Dehaene, Claire Sergent, Daniel Bliss & Panagiotia Theodoni
 - Klatzmann, U. et al. A dynamic bifurcation mechanism explains cortex-wide neural correlates of conscious access *Cell Reports* 10.1016/j.celrep.2025.115372 (2025)
- Spring 2019 **Master Research Project**, *NeuroSpin*, Paris, Modeling the Nonlinearity of Conscious Access
- Modeled the nonlinearity of neural responses in the dorsolateral prefrontal cortex found in van Vugt et al. data (2018) using a full-brain spiking network inspired by the model of Joglekar et al. (2018) and based on the connectivity matrix of Markov et al. (2012).
 - Supervised by: Stanislas Dehaene & Claire Sergent
 - Collaborators: Bram van Vugt, Jorge Mejias, Xiao-Jing Wang
 - Modeling tools: Brian and Alambic distributed computation framework
- Spring 2013 **Undergraduate Research Project**, *Columbia University*, New York, Modeling neuronal orientation tuning in V4
- Modeling the unsupervised specialization of neurons in cortical area V4, layer 2/3 in a spiking network based on physiological data from Hofer et al. (2011) using Spike-timing-dependent plasticity equations from Clopath et al. (2010).
 - Supervised by: Claudia Clopath
 - Modeling tools: Core Python

Teaching Experiences

- 2026 **Guest Lecturer**, *Université de Montréal*, Montréal
- Lecture: "Consciousness and Psychiatry"
- 2022–2024 **Teaching Assistant**, *École Normale Supérieure*, Paris
- Action, Decision Volition: NeuroCognitive Modeling of the evolution of the Prefrontal Executive System in Mammals.
 - Lecture: "Computational and neural bases of reinforcement learning in mammals"
 - Tutorials
 - Supervised by: Etienne Koechlin
- 2018–2022 **Tutor**, *OpenClassrooms*, Online
- Data science training for French job seekers

Entrepreneurial Experiences

- 2021–Present **Executive Director and co-founder**, *Mediterranean Association for Cognitive Science (MESEC)*
- Grassroots student organization in the field of consciousness science
 - €174,000 raised (grants from the Human Brain Project, the Templeton World Charity Foundation)
 - Organization of three academic events per year

- 2015–2018 **Growth & BI, OpenClassrooms**
- MOOC European leader, 3,500,000 monthly users
 - Redesign of the growth strategy
 - Built the Business Intelligence system from scratch
 - Scaled platform growth: +365% revenue, +2400% student growth
 - two fundraising rounds: €53M
- 2012–2014 **Executive Director and co-founder, Shapter**
- Online platform aggregating student reviews of academic courses.
 - 7 campuses, 2000+ students, €50,000 raised (Kima Ventures and French State)
 - Incubated at ParisTech Entrepreneur, Accelerated at TheFamily Fellowship
- 2012–2013 **AI Engineer Intern, Work4**
- Matching engine development (Python/C++)
 - Reshaped the distributed computing architecture (Celery)
 - Matching algorithms: Facebook profiles/ job offers

Grants, Scholarships, Fellowships & Awards

- 2026 **Prix Emmanuel Stip, Integration into clinical research, CAD 2,000**
- Department of Psychiatry and Addiction, Université de Montréal
- 2026 **Postdoctoral Fellowship, Impact+ Canada, CAD 140,000**
- Highly competitive national fellowship (2 years)
- 2025 **Research Grant, Freie Universität Berlin, EUR 15,000**
- START Funding Line
- 2025 **Scholarship, University of Bristol, GBP 3,000**
- PhD Engineering Mathematics program (visiting)
- 2024 **Grant, European Society for Cognitive Psychology, EUR 15,000**
- Supporting MESEC Workshop 2026
- 2023 **Grant, Templeton World Charity Foundation, USD 50,000**
- MESEC core funding
- 2022 **Grant, Templeton World Charity Foundation, USD 100,000**
- MESEC core funding
- 2021 **Grant, Human Brain Project, EUR 9,000**
- Supporting MESEC first Workshop

Technical skills

Programming languages

Main language - Python

Other languages - Visual Basic, C/C++, MATLAB, R, SQL (MySQL, Redshift), LaTeX

Other programming skills - Distributed Computing, Algorithmic, Big Data, Neuroimaging analysis, PyTorch, AI-assisted coding workflows, agentic AI

Sense of humor

situational humor, running gag, running gag.

Extra activities

Art & Literature

- Authors: M. Kundera, A. Maalouf, I. Asimov, P. Verlaine
- Composers: Phil Glass, Ludovico Einaudi, Steve Reich
- Scientists: K. Gödel, J. Monod, T. Kuhn

Travel

- Travel around the Mediterranean Sea, from Marseille to the Middle East
- Silk Road, from Paris to Xi'an, no plane nor hotel

- Upstream along the Amazon, from the Atlantic to the Pacific
- Upstream along the Nile, from Alexandria to Khartoum

Strategy games

- Board games, video games, card games

Sports

- Trekking (Corsica, Nepal, Thailand, Norway, Burma, Armenia, Kyrgyzstan, Ecuador, Pyrenees)
- Bouldering (7c+ European Level)

Workshops & Conferences

- 2026 **Oral presentation**, *Models of Consciousness 7*, AMCS, Copenhagen, Denmark
- 2026 **Poster presentation**, *Annual Meeting*, OHBM, Bordeaux, France
- 2025 **Oral presentation**, *Annual Meeting*, ASSC, Heraklion, Greece
- 2023 **Short talk**, *MESEC Workshop*, Corsica, France
- 2022 **Poster presentation**, *Annual Meeting*, ASSC, Amsterdam
- 2021 **Online talk**, *Conscious Brain Concept*, Online
- 2021 **Online poster presentation**, *Annual Meeting*, SFN, Online
- 2020 **Oral presentation**, *Forum des Sciences Cognitives*, Paris, France

Publications

- 2026 **Science from the bottom up**
 - Dohnány*, Hirschhorn*, **Klatzmann***, Matic*, Signorelli*, Titeux* & Tolle* — co-first author
 - *Trends in Cognitive Sciences*, <https://doi.org/10.1016/j.tics.2026.08.001>
- 2026 **What biology can, and cannot, tell us about conscious AI**
 - Klatzmann & Doerig
 - *Trends Open*, <https://doi.org/10.1016/j.treopn.2026.07.012>
- 2025 **Spatial layout of visual specialization is shaped by competing default mode and sensory networks**
 - Klatzmann et al.
 - *bioRxiv*, <https://doi.org/10.1101/2025.09.08.674858>
- 2025 **A dynamic bifurcation mechanism explains cortex-wide neural correlates of conscious access**
 - **Klatzmann***, Froudish-Walsh* et al.
 - *Cell Reports*, <https://doi.org/10.1016/j.celrep.2025.115372>